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Quiz for Week 1

coach

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1. (**Special note:** Below there are five questions for you to answer. Naturally there are many mathematical formulas and equations in the problem statements. However, due to some technical issues on Coursera, from time to time for some learners, some formulas and equations cannot be displayed correctly (while some others can). As there is no way for the instructing team to solve this issue, we post all the problem statements in a PDF file here. If unfortunately some formulas and equations are not readable for you, please use the PDF file to prepare your answers and then come back to Coursera to choose/fill in your answers.) **1 / 1 point**

Consider the following linear program

$$\begin{array}{rl} \max & x_1 + 3x_2 \\ \text{s.t.} & -x_1 + x_2 \le 3 \\ & -x_1 + 2x_2 \le 8 \\ & 3x_1 + x_2 \le 18 \\ & x_i \ge 0 \quad \forall i = 1, 2. \end{array}$$

What is an optimal solution to this linear program?

- (●) $(4, 6)$.
- ( ) $(6, 0)$.
- ( ) $(0, 3)$.
- ( ) $(0, 0)$.
- ( ) None of the above.

Correct

2. Continue from the previous question. Let the slack variables for constraints 1, 2, and 3 be x_3, x_4, and x_5, respectively. Consider the basis $B = (x_2, x_4, x_5)$. What is the corresponding basic feasible solution? **1 / 1 point**

- ( ) There is no corresponding basic feasible solution.
- ( ) $(0, 0, 3, 8, 18)$.
- ( ) $(0, 3, 2, 0, 15)$.
- ( ) $(0, 2, 3, 0, 15)$.
- (●) None of the above.

Correct

3. Continue from the previous question. Let $N = (x_1, x_3)$ be the set of nonbasic variables. What is the matrix A_N ? **1 / 1 point**

- ( ) $\begin{bmatrix} -1 & 1 \\ -1 & 2 \\ 3 & 1 \end{bmatrix}$.
- (●) $\begin{bmatrix} -1 & 1 \\ -1 & 0 \\ 3 & 0 \end{bmatrix}$.
- ( ) $\begin{bmatrix} -1 & 1 \\ 0 & 1 \\ 3 & 0 \end{bmatrix}$.
- ( ) $\begin{bmatrix} -1 & 1 \\ 1 & -2 \\ 4 & -1 \end{bmatrix}$
- ( ) None of the above.

Correct

4. Continue from the previous question. What is the transpose of the reduced cost vector $\bar{c}_N^T$? **1 / 1 point**

- ( ) $\begin{bmatrix} 1 & 0 \end{bmatrix}$
- ( ) $\begin{bmatrix} 0 & 1 \end{bmatrix}$.
- (●) $\begin{bmatrix} -4 & 3 \end{bmatrix}$.
- ( ) $\begin{bmatrix} 3 & -4 \end{bmatrix}$.
- ( ) None of the above.

Correct

5. Continue from the previous question. According to the reduced costs and ratio test, which variable will be the leaving variable at this basis? **1 / 1 point**

- ( ) x_1.
- ( ) x_2.
- (●) x_4.
- ( ) x_5.
- ( ) None of the above.

Correct